



The Green Village

H₂Integration

Into A Residential
Energy Hub

Daan Schat

Introduction

The Green Village Energy Hub

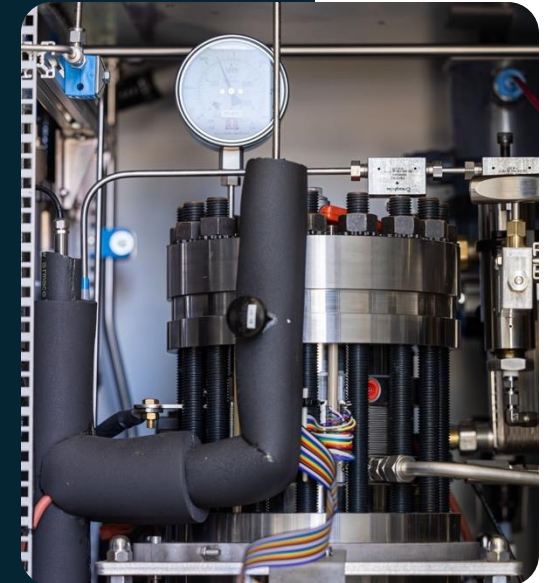
- Renewable energy generation
- Short term energy storage in battery
- Full hydrogen cycle:
 - Hydrogen production
 - Hydrogen storage
 - Electricity generation from hydrogen



Introduction

The Green Village Energy Hub

- Supplies energy to a 1-person house
- Designed to be grid independent



- Battery: BYD LVL LFP 15.4 kWh
- Solar Panels: DEMGC 12 X 445 W_p
- Electrolyser: 2.4 kW AEM Enapter EL2.1
- Compressor: HyET electrochemical comp.
- Hydrogen storage: 53X50L ~48 kg @270 Bar
- Fuel Cell: PEM Nedstack FCS 7-XXL 6.8 kw



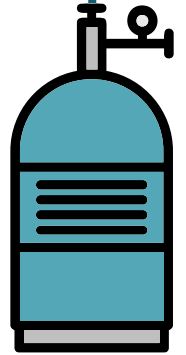
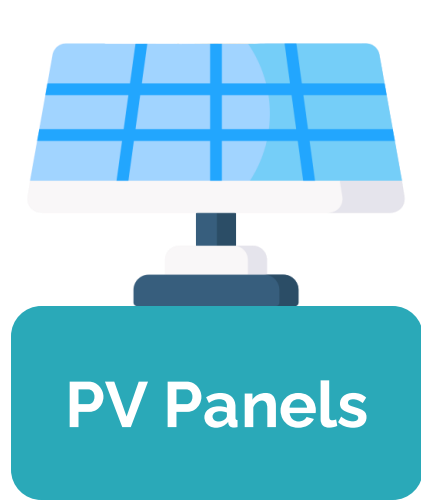
Introduction

System size

Components sizing

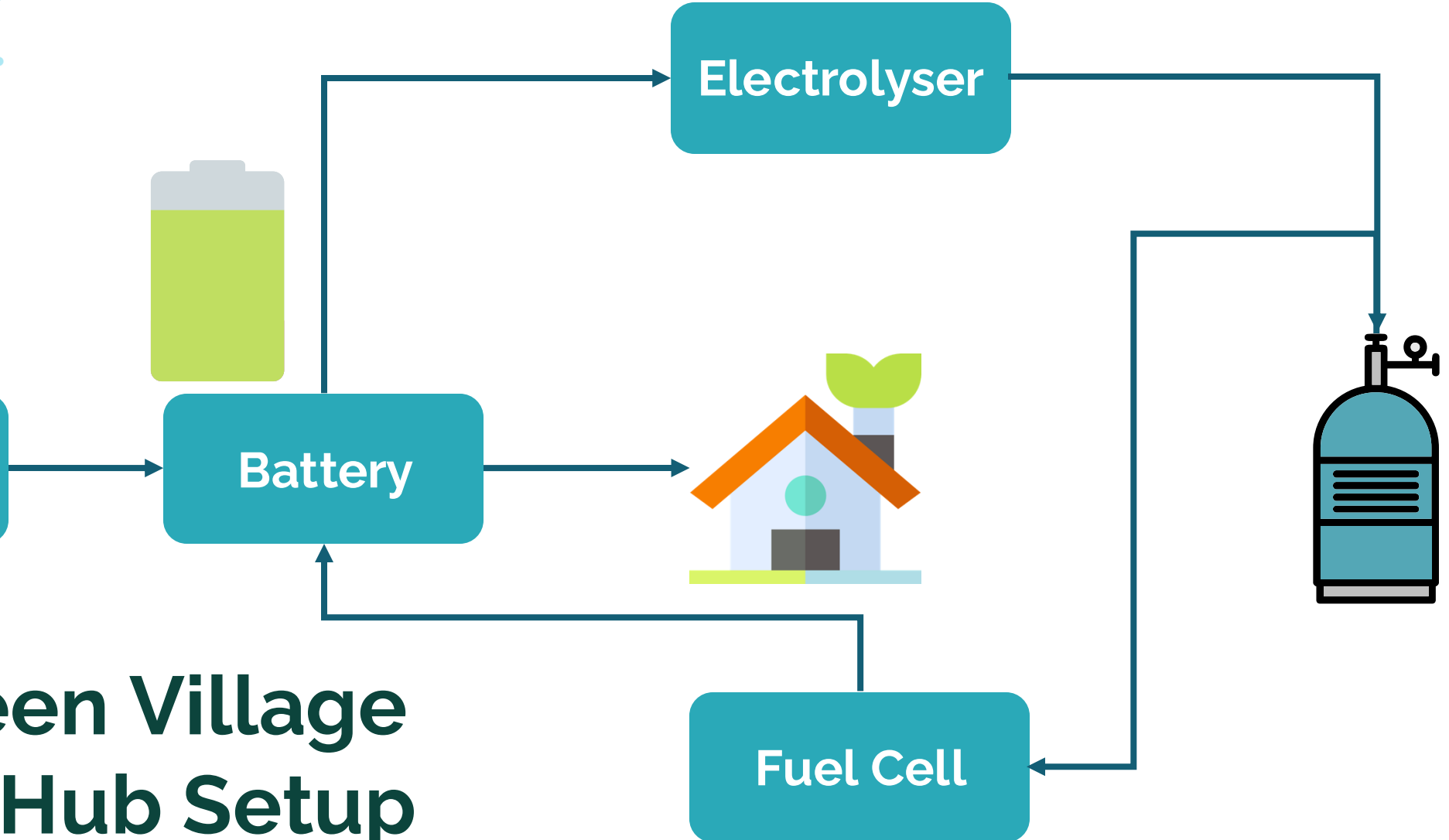
The Green Village energy hub is scaled to accommodate the energy consumption of a one-person fully electrified home.





Introduction

The Green Village Energy Hub Setup



Electrolyser

Often operational during summer

Starts if all conditions are met:

- PV Power > 1 kW
- And time before  12:00
- And battery SoC > 80%

Stops:

- Battery SoC < 65%

Introduction

The Green Village Energy Hub Setup

Fuel Cell

Often operational during winter

Starts:

- If Battery SoC < 25% (at any time)
- During Winter If:
 - SoC < 40% before  13:00
 - SoC < 50% before  14:00
 - SoC < 60% before  15:00
 - SoC < 70% before  16:00
 - SoC < 80% before  17:00

Stops:

- If battery SoC > 90%



DATA **RESULTS**





Key findings

Results

- Electrolyser efficiency & behaviour
- Fuel cell efficiency
- Hydrogen buffer size
- Balance of Plant consumption
- Efficiency optimisation: avoid reaching 100% SoC



64.8%

Best electrolyser efficiency
At 70% capacity

51.5%

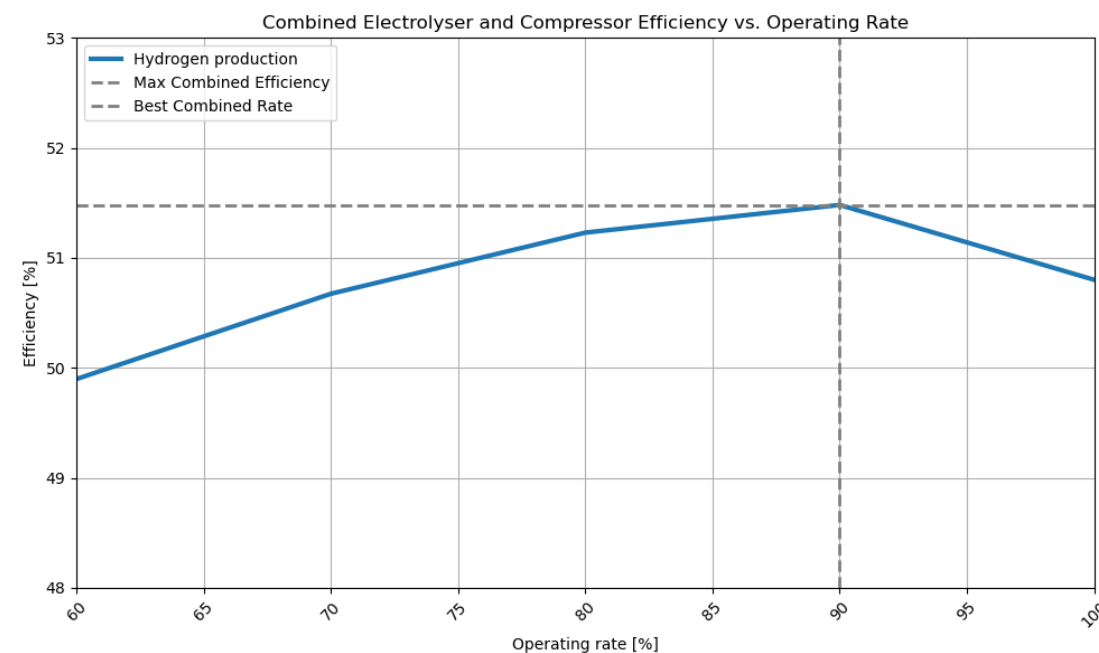
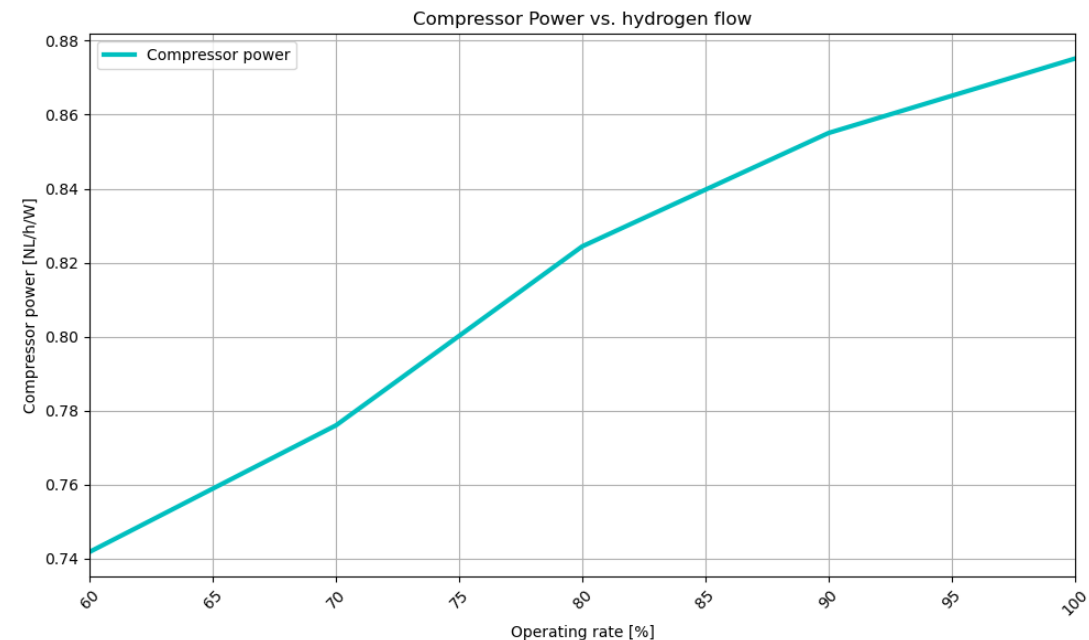
Best combined electrolyser
& compressor efficiency
At 90% capacity

90%

Best operational rate

Key findings

Electrolyser & compressor efficiency



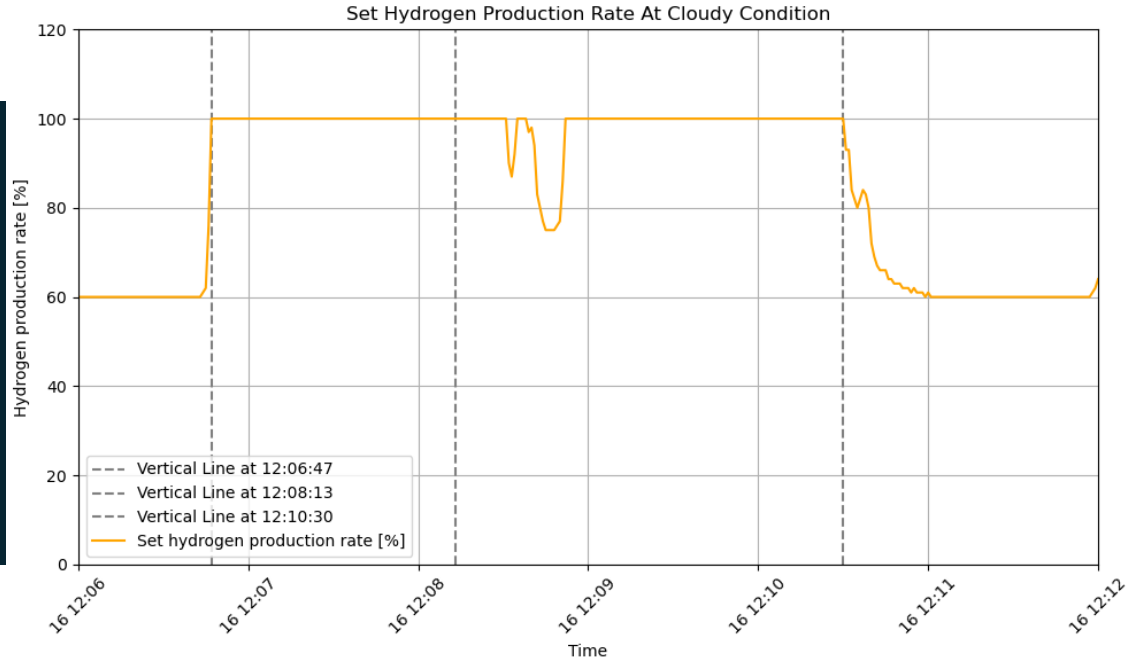
Key findings

Electrolyser & Compressor

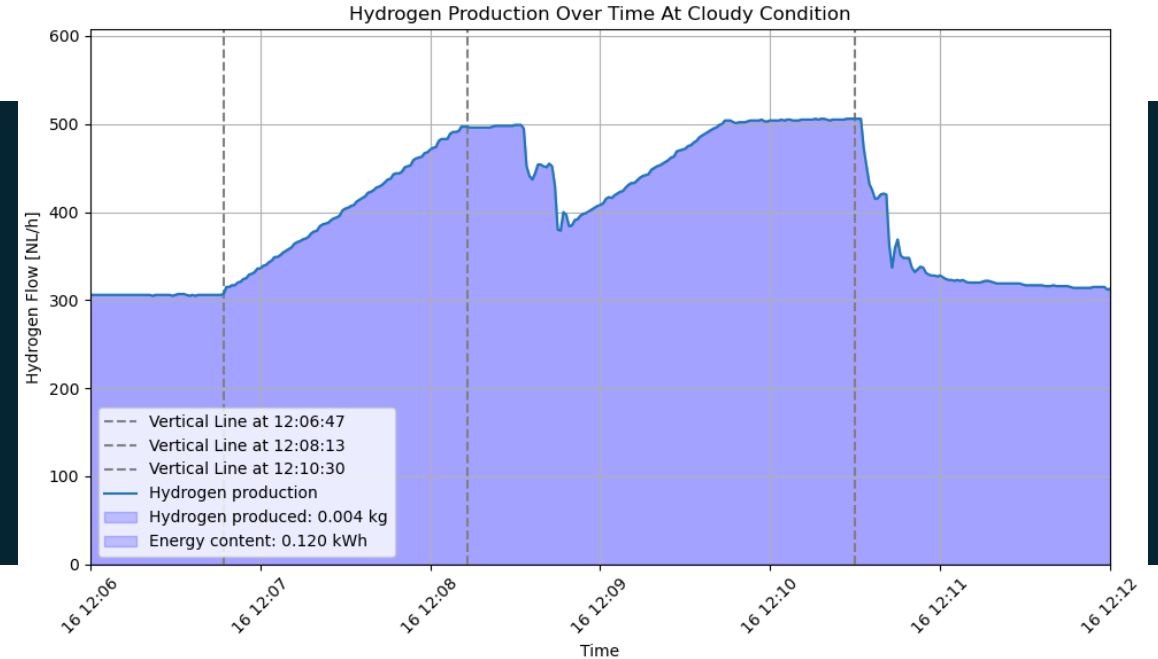
Following PV power

When the electrolyser was set to follow the PV power, we could see that the production rate is slow to climb, and quick to fall. Making this way of usage undesirable.

- Electrolyser prod. rate climbs slowly
- Electrolyser prod. Rate falls quickly
- More production was possible



Hydrogen production rate set point



Actual production rate

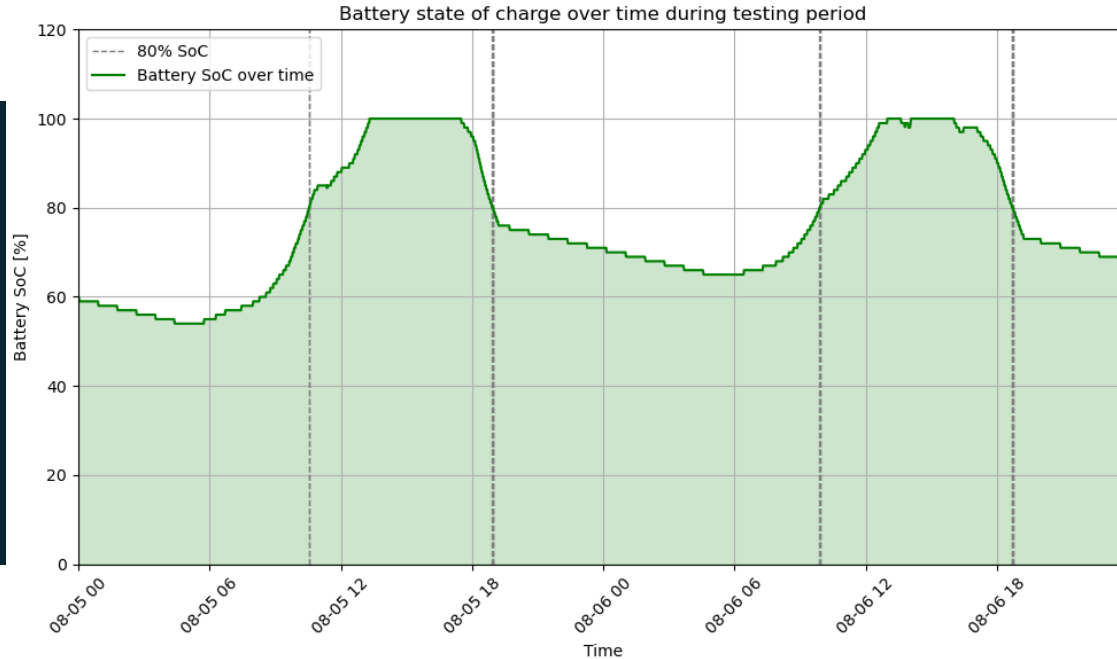
Key findings

Electrolyser & Compressor

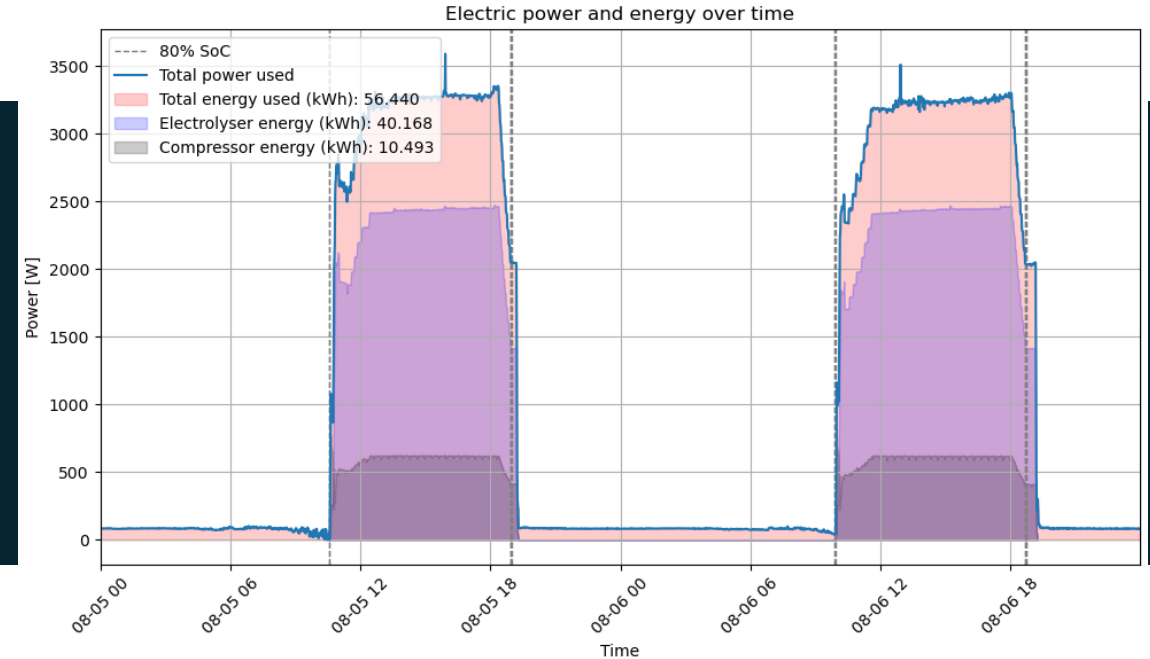
Intended use

When the electrolyser was used in the intended use-case, we could see that the SoC reached 100%, wasting the excess of electricity. This means the electrolyser could've operated for longer, making more hydrogen.

- 100 % SoC reached, so grid exchange necessary
- More hydrogen could be produced
- System efficiency of 44.6%: PV inverter to hydrogen, incl. BoP*



Battery SoC



Power consumption

Fuel cell efficiency results:

38.7%*

Stack power to inverter
efficiency (incl. BoP)

20.9%

Hydrogen to inverter
efficiency (incl. BoP)

Fuel cell results:

2380 W

Stack output power

1800 W

Inverter output power

16.5%

Power converter losses

+375 W

BoP power consumption

Key findings

Fuel cell results

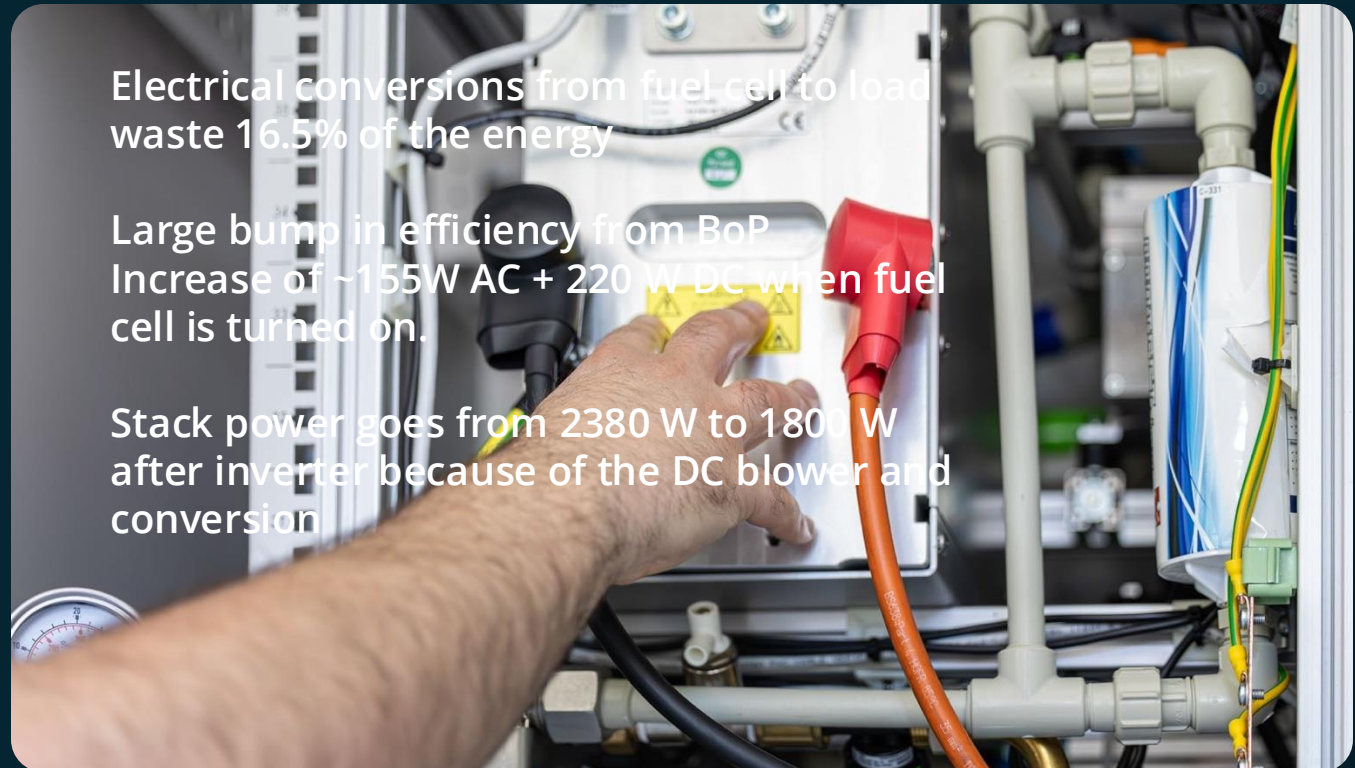
Data analysis

The Green Village energy hub is scaled to accommodate the energy consumption of a one-person fully electrified home.

Electrical conversions from fuel cell to load
waste 16.5% of the energy

Large bump in efficiency from BoP
Increase of ~155W AC + 220 W DC when fuel
cell is turned on.

Stack power goes from 2380 W to 1800 W
after inverter because of the DC blower and
conversion

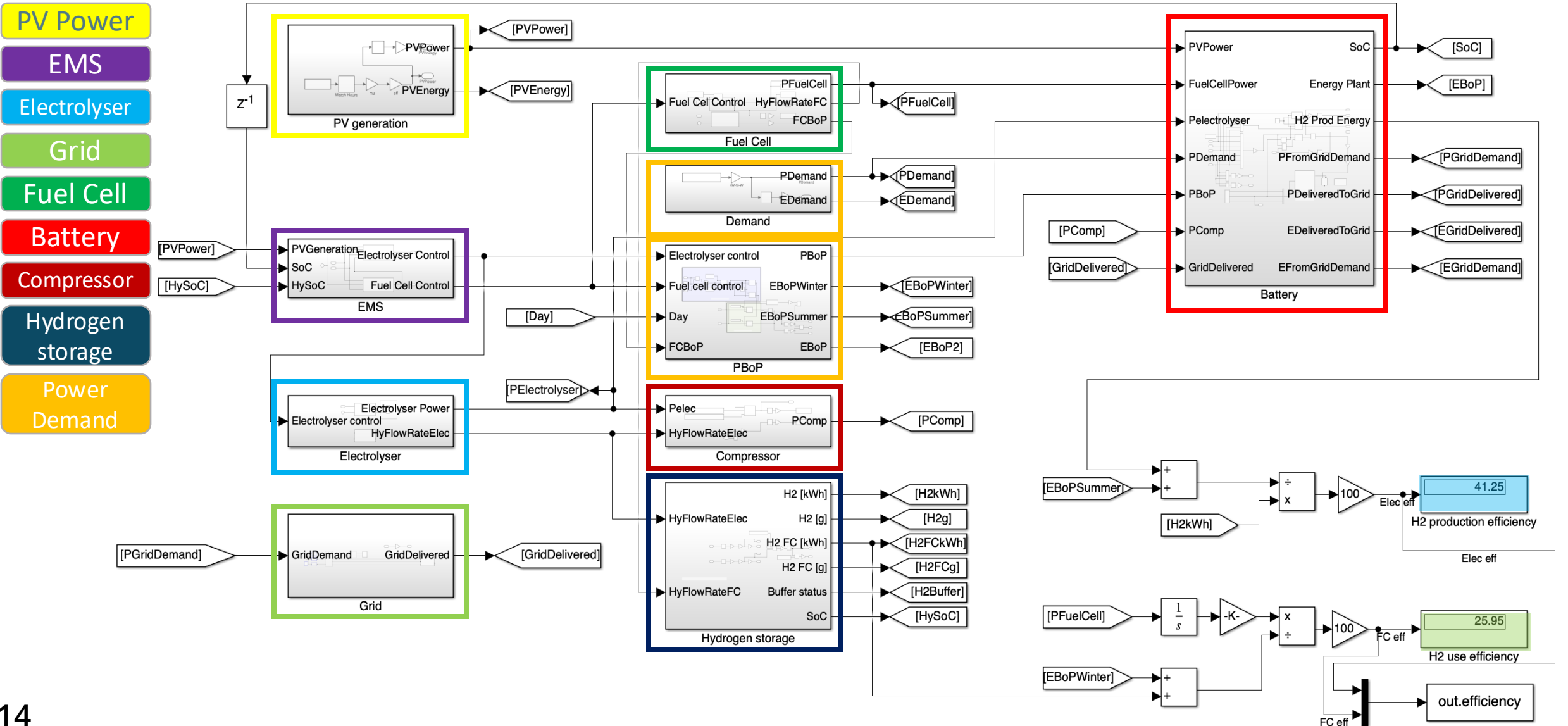


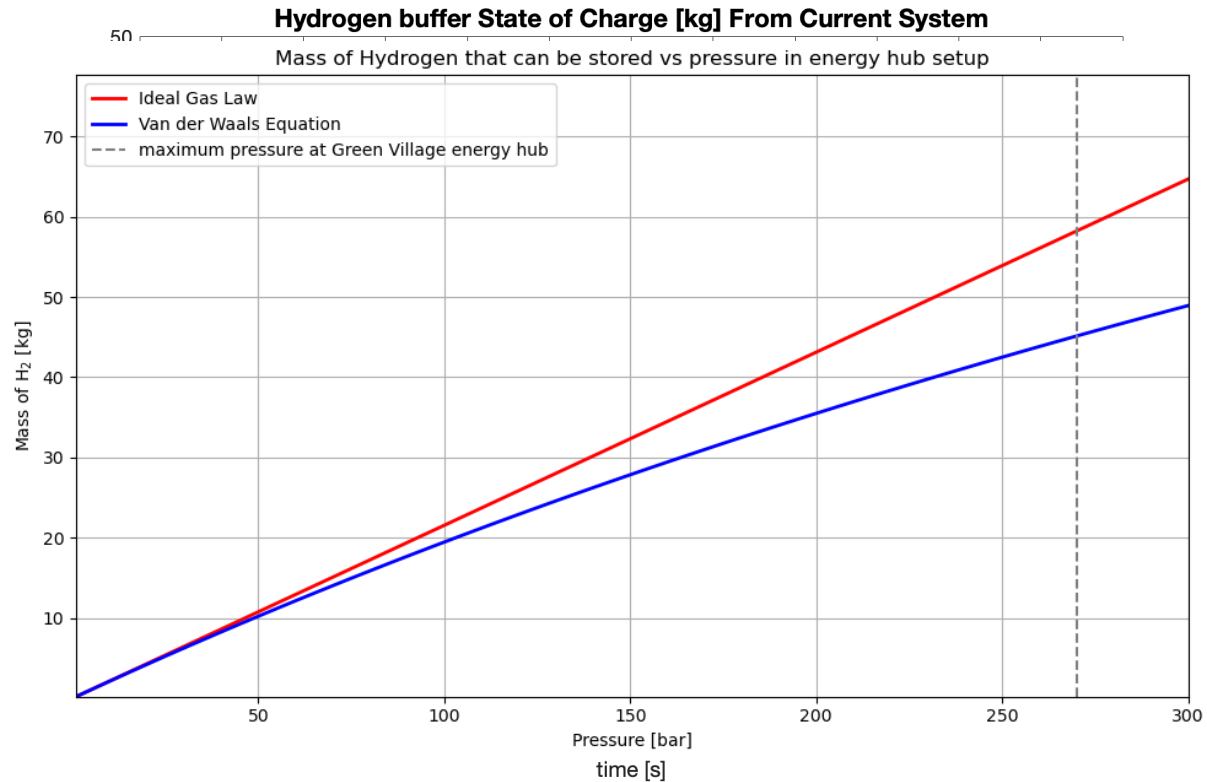


SIMULATION **RESULTS**



Simulink Digital Twin





- Large mismatch in start and end of year buffer status
- Total hydrogen use: 57.4 kg*
- Total hydrogen production: 33.5 kg

* Starting buffer state is 24 kg (50%)

- Changing the EMS goes a long way for improvement
- Allowing the electrolyser to turn on before 14:00 instead of before 12:00 increased production by 35% and increased efficiency by 1.3%

Key findings

Simulations

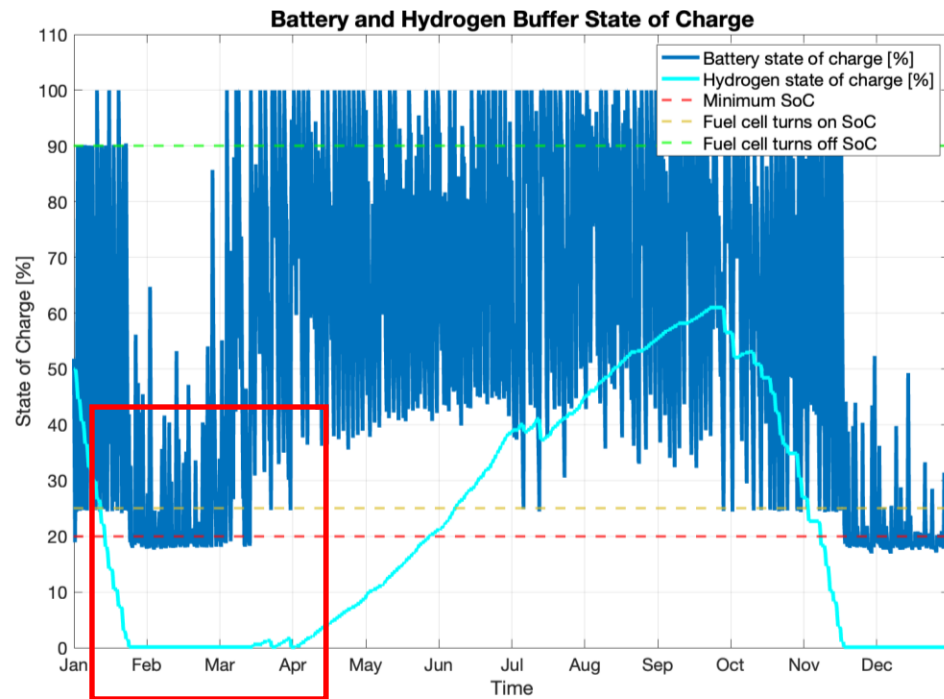
System modelling: Current system

The simulations shown in this slide are the results for the current EMS setup and currently integrated system

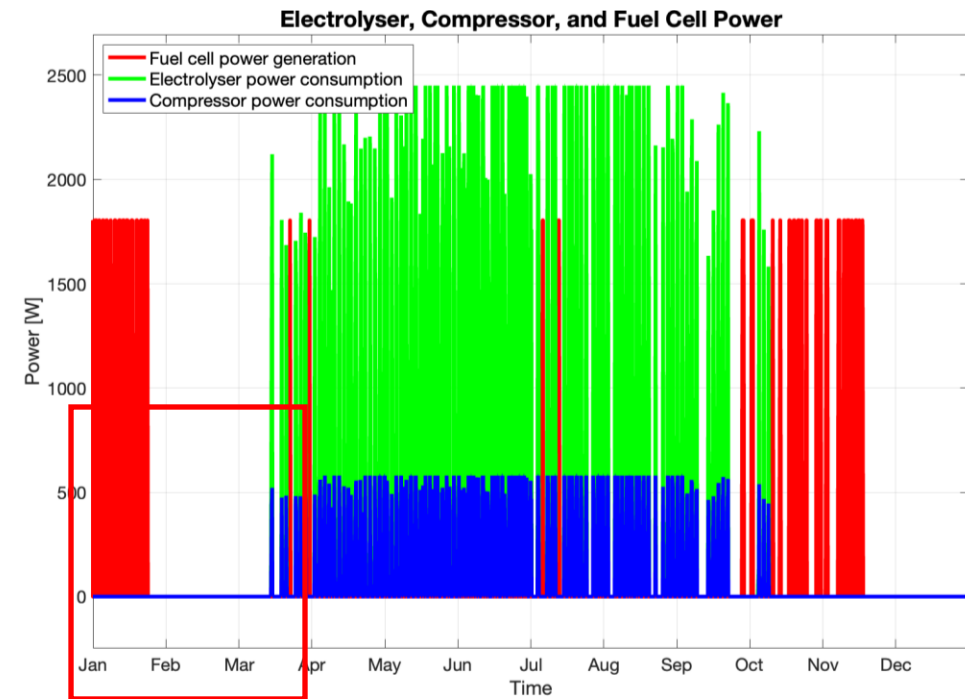
Key findings

State of Charge

- 100 % SoC reached, so grid exchange necessary
- 20 % SoC reached, so grid exchange necessary
- System efficiency of 44.6%: PV inverter to hydrogen, incl. BoP



State of Charge

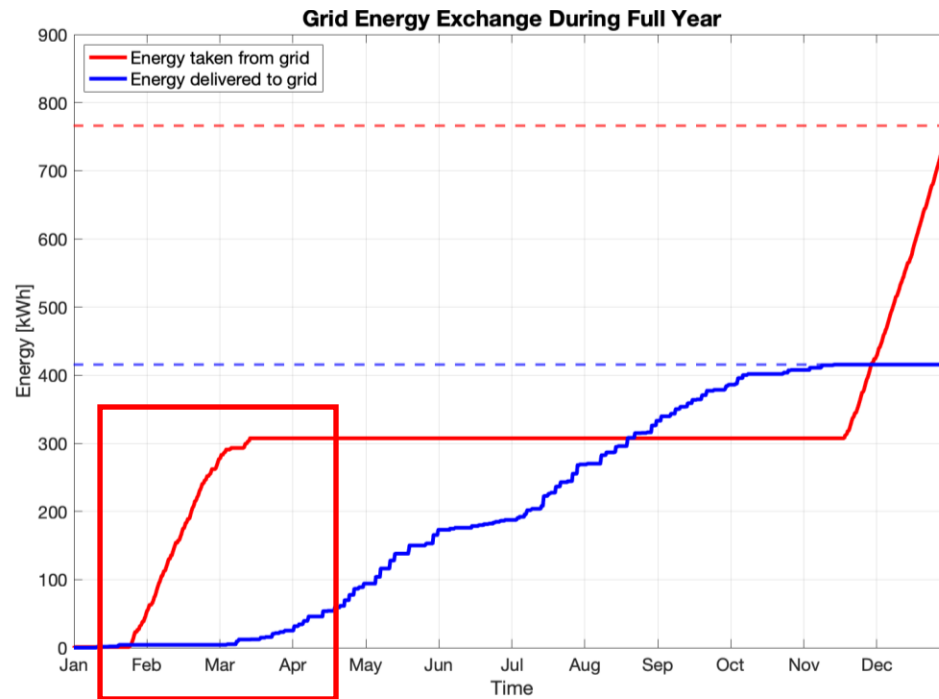


Component power

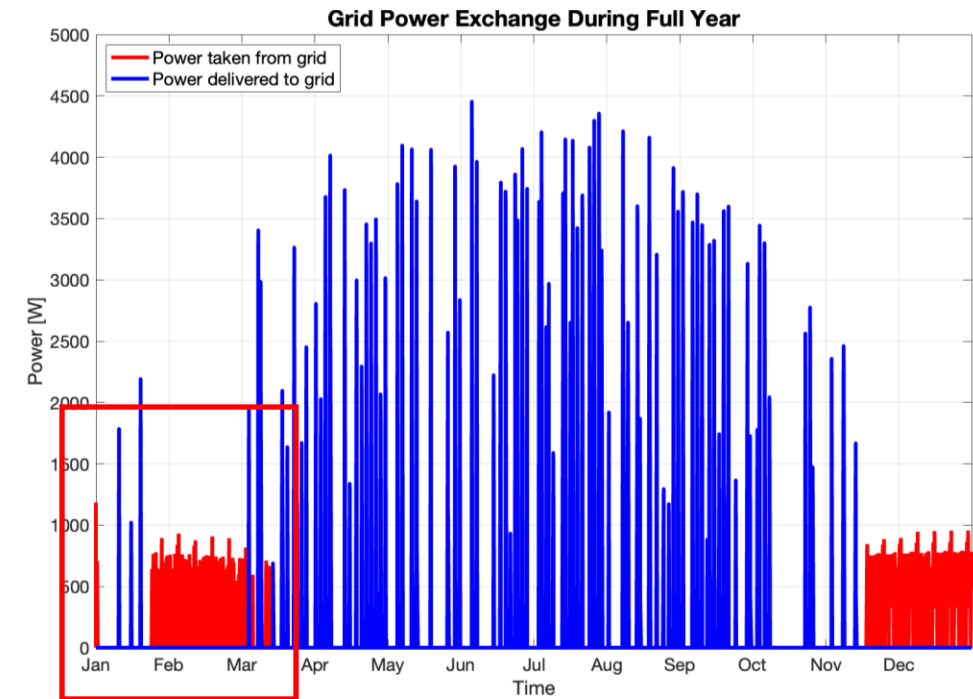
Key findings

Grid Exchange

- Total PV energy yield is 5100 kWh
- 8% Of PV energy goes to the grid
- 19.5% Of the demand is pulled from the grid



Energy to and from grid



Power to and from grid



COMPARING **RESULTS**

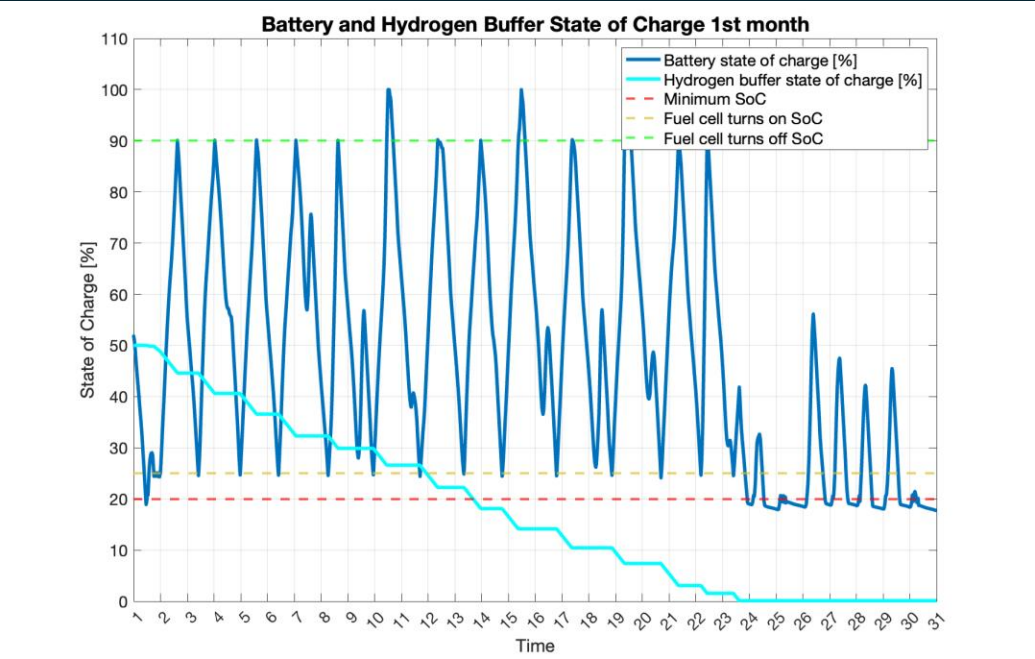
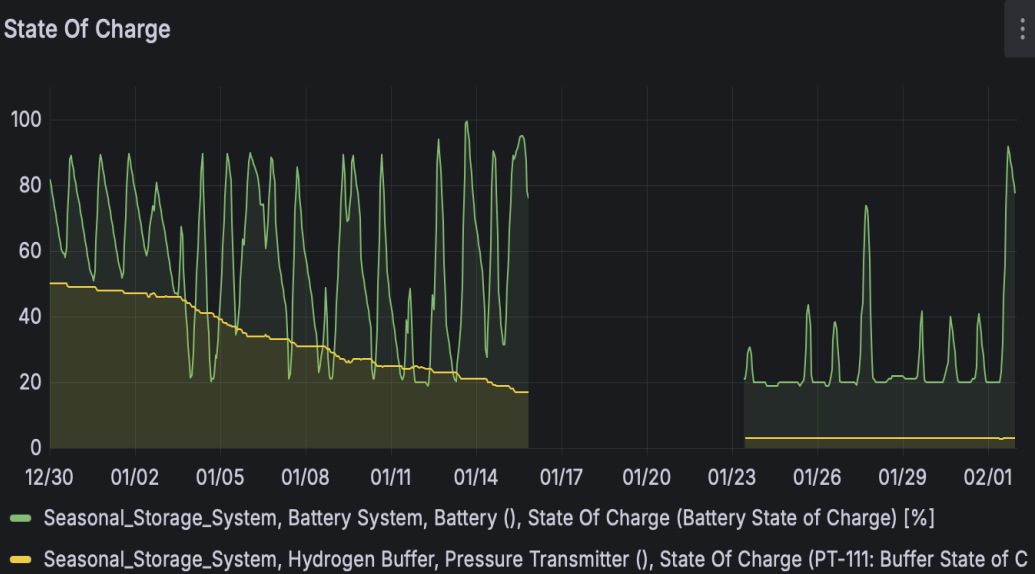


Conclusion

	Data results	Simulation results
Electrolyser efficiency incl. BoP	44.6%*	41.3%**
Fuel cell efficiency incl. BoP	20.9%	25.9%

- Changing the EMS greatly influences efficiency
- Combining the energy hub with heat improves efficiency
- Digital twin is an easy and accurate way to simulate changes to the system efficiently

Hydrogen buffer 1st month of the year:



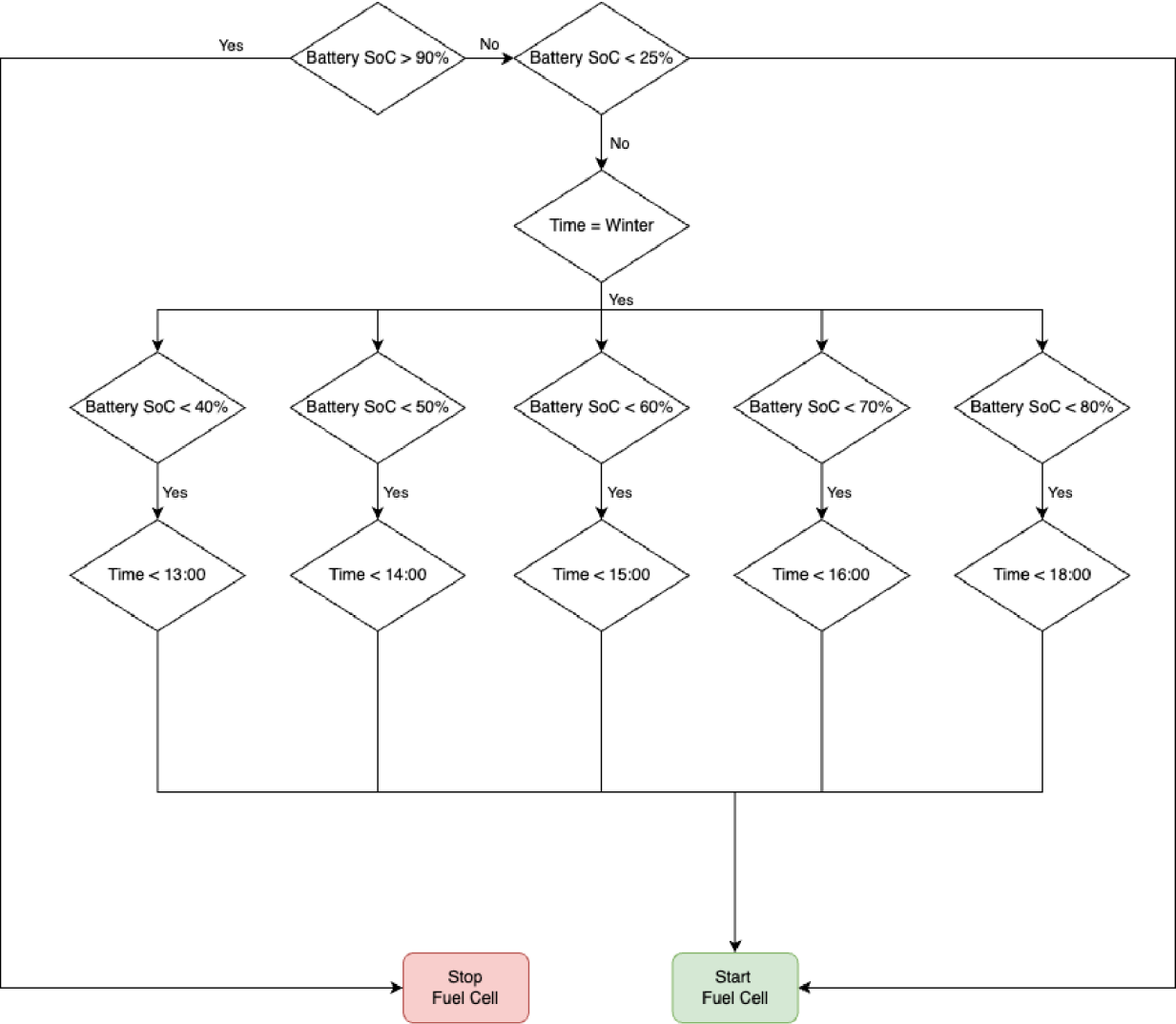
* Calculated over sunny days only
** Calculated over an entire year

Thanks For Your Attention

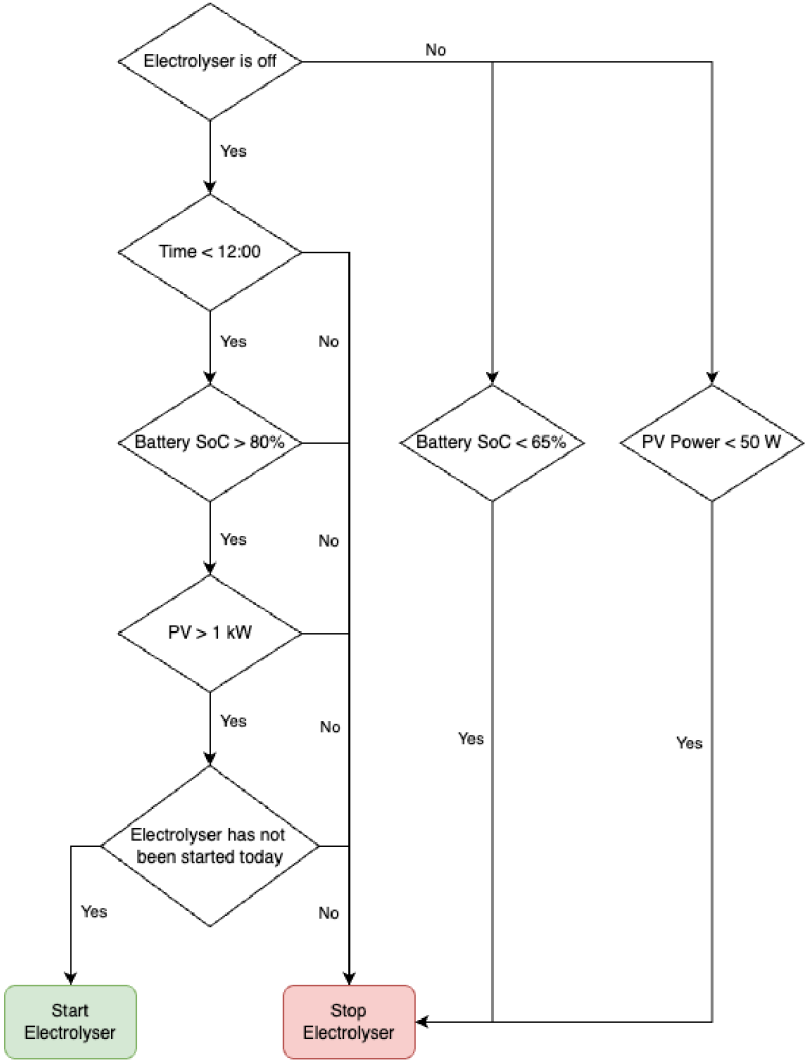
THANK YOU

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Fuel Cell



Electrolyser



Model

- Original idea of hydrogen capacity: 60 kg
- Calculated hydrogen capacity (vdw): 48 kg
- Difference of 400 kWh_{ch}
- Hydrogen buffer went from 55% to 17% in 19 days

During the winter simulations:

802 kWh Generated PV energy

1078 kWh Energy demand

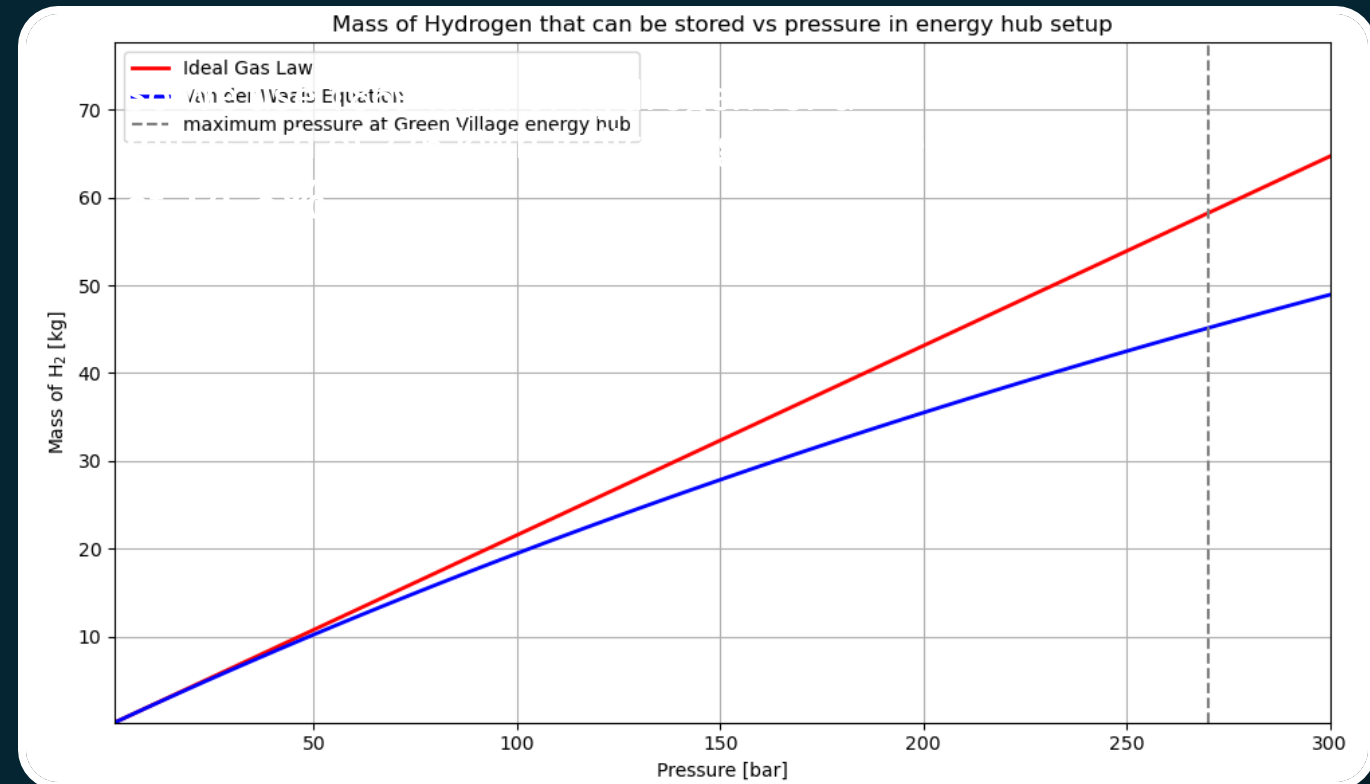
56.09 kg Hydrogen used

1869 kWh Hydrogen used

Key findings

Simulations

Hydrogen capacity



Appendix

System

